

Mengxi Wu

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Research Interests

Developing efficient methods for large language model pre-training, with a focus on training dynamics, hyperparameter transfer, and optimizers for large-scale training.

Education

University of Southern California <i>Doctor of Philosophy, Computer Science</i>	Expected 2027.12
University of Southern California <i>Master of Arts, Pure Mathematics</i>	Expected 2027.5
New York University <i>Master of Science, Computer Science</i>	2021.5
University of Michigan, Ann Arbor <i>Bachelor of Science in Engineering, Electrical Engineering</i>	2019.5

Experience

Honeywell <i>AI PhD Intern</i> Developed AI models to solve targeted business problems.	2026.5 - 2026.8
Huawei <i>Machine Learning Engineer</i> Worked on object detection in construction sites.	2021.9 - 2022.5

Publications

(* indicates equal contribution.)

- [7] **Towards Self-Adaptive Learning: Continual Learning under Harsh Conditions.**
Roozbeh Razavi-Far, Ehsan Hallaji¹, Alireza Fathalizadeh, Mengxi Wu, Mohammad Rostami.
[Submitted](#)
- [6] **Gecko: An Efficient Neural Architecture Inherently Processing Sequences with Arbitrary Lengths.**
Xuezhe Ma^{*}, Shicheng Wen^{*}, Linghao Jin^{*}, Bilge Acun^{*}, Ruihang Lai^{*}, Bohan Hou, Will Lin, Hao Zhang, Songlin Yang, Ryan Lee, Mengxi Wu, Jonathan May, Luke Zettlemoyer, Carole-Jean Wu.
[Submitted](#)
- [5] **GQA- μ P: The Maximal Parameterization Update for Grouped Query Attention and Fully Sharded Data Parallel.**
Kyle R. Chickering^{*}, Huijuan Wang^{*}, Mengxi Wu^{*}, Alexander Moreno, Muhao Chen, Xuezhe Ma, Daria Soboleva, Joel Hestness, Zhengzhong Liu, Eric P. Xing.
[Submitted](#)

- [4] **Curvature Diversity-Driven Deformation and Domain Alignment for Point Cloud.**
Mengxi Wu, Hao Huang, Yi Fang, Mohammad Rostami.
[Transactions on Machine Learning Research 2025](#)
- [3] **Graph Harmony: Denoising and Nuclear-Norm Wasserstein Adaptation for Enhanced Domain Transfer in Graph-Structured Data.**
Mengxi Wu, Mohammad Rostami.
[Transactions on Machine Learning Research 2024](#)
- [2] **Streaming Approach to In Situ Selection of Key Time Steps for Time-Varying Volume Data.**
Mengxi Wu, Yi-Jen Chiang, Christopher Musco.
[Eurographics/IEEE Conference on Visualization 2022](#)
- [1] **3D Point Cloud Completion with Geometric-Aware Adversarial Augmentation.**
Mengxi Wu, Hao Huang, Yi Fang.
[International Conference on Pattern Recognition 2022](#)

Teaching

University of Southern California

CSCI 544 Applied Natural Language Processing, *Teaching Assistant*

CSCI 570 Analysis of Algorithms, *Teaching Assistant*

New York University

ECE-GY 9123 Deep Learning, *Teaching Assistant*

Services

Conference Reviewing

EMNLP, ICLR, ICML, NeurIPS

Journal Reviewing

Transactions on Machine Learning Research

Skills

Programming Languages

C/C++, Python, Java, R, MATLAB, Swift

Libraries and Tools

PyTorch, Tensorflow, PySpark, Hadoop